

## Research Interests

My research interests lie in **Computer Vision, Multimodal Learning, Vision-Language Models (VLMs), and Trustworthy AI**. I am interested in developing **reliable and interpretable AI systems** that bridge visual perception and multimodal reasoning for real-world applications. My current research focuses on:

- (1) **Efficient visual understanding**, including object detection, scene understanding, and lightweight architectures for real-time inference;
- (2) **Multimodal learning**, integrating visual, linguistic, and contextual information for video understanding and video-to-audio generation;
- (3) **Medical image analysis**, developing robust deep learning models for computer-aided diagnosis and trustworthy clinical AI systems;
- (4) **Trustworthy AI**, with emphasis on robustness, interpretability, privacy-preserving machine learning, and the safe deployment of vision-language models.

## Education

### Sogang University (Transferred 2024)

Seoul, South Korea

B.S. IN COMPUTER SCIENCE

Expected Aug. 2026

- Relevant Coursework: Deep Learning, Machine Learning, Big Data Computing

### Hankuk University of Foreign Studies (HUFS)

Seoul, South Korea

INFORMATION & COMMUNICATION ENGINEERING

2020 – 2023

- Transferred to Sogang University in 2024

## Projects

### MULTI-TRACK VIDEO-TO-AUDIO DESCRIPTION (4 MEMBERS)

Mar 2026 – Jun 2026

- Developed an end-to-end **Vision-Language Model (VLM)** pipeline that analyzes videos, generates structured multi-track audio (dialogue, sound effects, music, and ambience), and synthesizes the final mixed video.
- Designed a reference-based evaluation framework for improving semantic consistency between visual understanding and generated audio descriptions.
- **Gold Prize (1st Place)**, Sogang University Capstone Design Competition (2026).
- **Paper accepted to KCC 2026** (Multi-Track Video-to-Audio Description).  
GitHub: [github.com/sogang-v2a/v2a\\_inspect](https://github.com/sogang-v2a/v2a_inspect)

### CHEST X-RAY MULTI-DISEASE DETECTION SYSTEM (CXR-CAD) (5 MEMBERS, TEAM LEADER)

Mar 2026 – Jun 2026

- Developed a multi-label chest X-ray disease diagnosis system using deep learning with **Focal Loss** and **Grad-CAM**.
- Designed a deployment-oriented evaluation framework considering robustness, calibration, and distribution shifts for trustworthy clinical AI.
- **Encouragement Prize**, Sogang University Capstone Design Competition (2026)
- **Paper accepted to KCC 2026**.  
GitHub: [github.com/sogang-cxr-cad/CXR-CAD](https://github.com/sogang-cxr-cad/CXR-CAD)

### RAG-BASED AI CRM MANAGER (5 MEMBERS, TEAM LEADER)

Sep 2025 – Dec 2025

- Built a **Multimodal RAG** system on GCP using GPT-4o, ChromaDB, FastAPI, and PostgreSQL for customer support automation.
- Developed autonomous customer-service and BI reporting agents through a full-stack architecture.  
GitHub: [github.com/suhwantcha/CS-Agent](https://github.com/suhwantcha/CS-Agent)

### FLIGHT DELAY PREDICTION (4 MEMBERS)

Nov 2025

- Built a flight arrival delay prediction model using an **XGBoost + LightGBM** ensemble with statistical feature engineering.
- **Excellence Award**, 2025 Chung-Ang × Sogang University Joint Datathon.  
GitHub: [github.com/suhwantcha/Flight-Delay-Prediction](https://github.com/suhwantcha/Flight-Delay-Prediction)

### PINTOS OPERATING SYSTEM

Sep 2025 – Dec 2025

- Implemented user programs, priority scheduling, file systems, and synchronization mechanisms in the Pintos educational operating system using C.  
GitHub: [github.com/suhwantcha/OS-Pintos-Project](https://github.com/suhwantcha/OS-Pintos-Project)

### YOLO-PSYCHO-ANALYSIS

Apr 2025 – May 2025

- Developed a YOLOv8-based psychological assessment system that analyzes hand-drawn house sketches based on HTP theory.  
GitHub: [github.com/suhwantcha/YOLO-Psycho-Analysis](https://github.com/suhwantcha/YOLO-Psycho-Analysis)

## Experience

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| <b>Teaching Assistant, Basic Artificial Intelligence Programming</b><br>SOGANG UNIVERSITY <ul style="list-style-type: none"><li>Assisted Python programming labs, answered student questions, graded assignments, and managed course administration.</li></ul>                                                                                                                                                                            | Seoul, South Korea<br>Mar 2026 – Jun 2026   |
| <b>Teaching Assistant, Understanding Artificial Intelligence</b><br>SOGANG UNIVERSITY <ul style="list-style-type: none"><li>Supported an introductory AI course for 50+ students through Q&amp;A sessions, grading, and course management.</li><li>Conducted PyTorch-based laboratory sessions covering CNNs and Transformer models.</li></ul>                                                                                            | Seoul, South Korea<br>Sep 2025 – Dec 2025   |
| <b>Teaching Assistant, Computer Architecture</b><br>SOGANG UNIVERSITY <ul style="list-style-type: none"><li>Assisted students with computer architecture coursework and laboratory exercises.</li></ul>                                                                                                                                                                                                                                   | Seoul, South Korea<br>Mar 2025 – Jun 2025   |
| <b>Member &amp; Mentor, Parrot Data Science Club</b><br>SOGANG UNIVERSITY <ul style="list-style-type: none"><li>Participated in machine learning and computer vision study groups, collaborating on data science projects.</li><li>Mentored five junior members in data analysis and CNN-based computer vision projects.</li><li>Organized Kaggle-style machine learning competitions and technical workshops for club members.</li></ul> | Seoul, South Korea<br>Sep 2024 – Present    |
| <b>Teaching Assistant, English &amp; Mathematics</b><br>MEGASTUDY RUSSEL & YESUM ACADEMY <ul style="list-style-type: none"><li>Tutored middle and high school students while assisting classroom instruction in mathematics and English.</li></ul>                                                                                                                                                                                        | Seoul, South Korea<br>Feb 2024 – Present    |
| <b>Satellite Operation Squad Leader</b><br>REPUBLIC OF KOREA ARMY <ul style="list-style-type: none"><li>Operated military satellite communication systems to ensure reliable communications across military units.</li><li>Led a team of seven personnel responsible for satellite equipment operation, maintenance, and inspection.</li></ul>                                                                                            | Daejeon, South Korea<br>Jan 2022 – Jul 2023 |

## Activities

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| <b>Autonomous Taxi Design Project</b><br>SOGANG UNIVERSITY <ul style="list-style-type: none"><li>Designed an autonomous taxi system with emphasis on AI-based navigation, perception, and safety.</li><li>Proposed cost-effective sensor fusion strategies for reliable environmental perception.</li></ul>                                                         | Seoul, South Korea<br>Jul 2025 – Aug 2025 |
| <b>Runner-up, AI Data Analysis Runnerthon</b><br>SOGANG UNIVERSITY <ul style="list-style-type: none"><li>Achieved <b>2nd place</b> among 15 participating teams.</li><li>Developed an XGBoost-based real estate price prediction model through feature engineering and ensemble learning.</li></ul>                                                                 | Seoul, South Korea<br>Jan 2025 – Feb 2025 |
| <b>Software and AI Competency Enhancement Training Program</b><br>REPUBLIC OF KOREA ARMY <ul style="list-style-type: none"><li>Completed an intensive software and AI training program covering programming and machine learning fundamentals.</li><li>Developed a machine learning-based weather forecasting model using real-world meteorological data.</li></ul> | South Korea<br>Jan 2023 – May 2023        |
| <b>English Training Program (R-EIP)</b><br>HANKUK UNIVERSITY OF FOREIGN STUDIES (HUFS) <ul style="list-style-type: none"><li>Completed an intensive academic English program focused on communication and presentation skills.</li></ul>                                                                                                                            | Seoul, South Korea<br>Oct 2021 – Dec 2021 |

## Technical Skills

- Programming** Python, C/C++, SQL, R
- Machine Learning** PyTorch, TensorFlow, scikit-learn, Hugging Face
- Data & Computer Vision** NumPy, Pandas, OpenCV, Matplotlib
- Backend & Database** FastAPI, MySQL, ChromaDB
- Tools** Git, Linux, Docker, VS Code, Jupyter Notebook, Google Colab

## Languages

- Korean:** Native
- English:** Fluent (TOEFL iBT 100, Feb 2026)
- Japanese:** Basic